

DIGITAL GHG INVENTORY SYSTEM

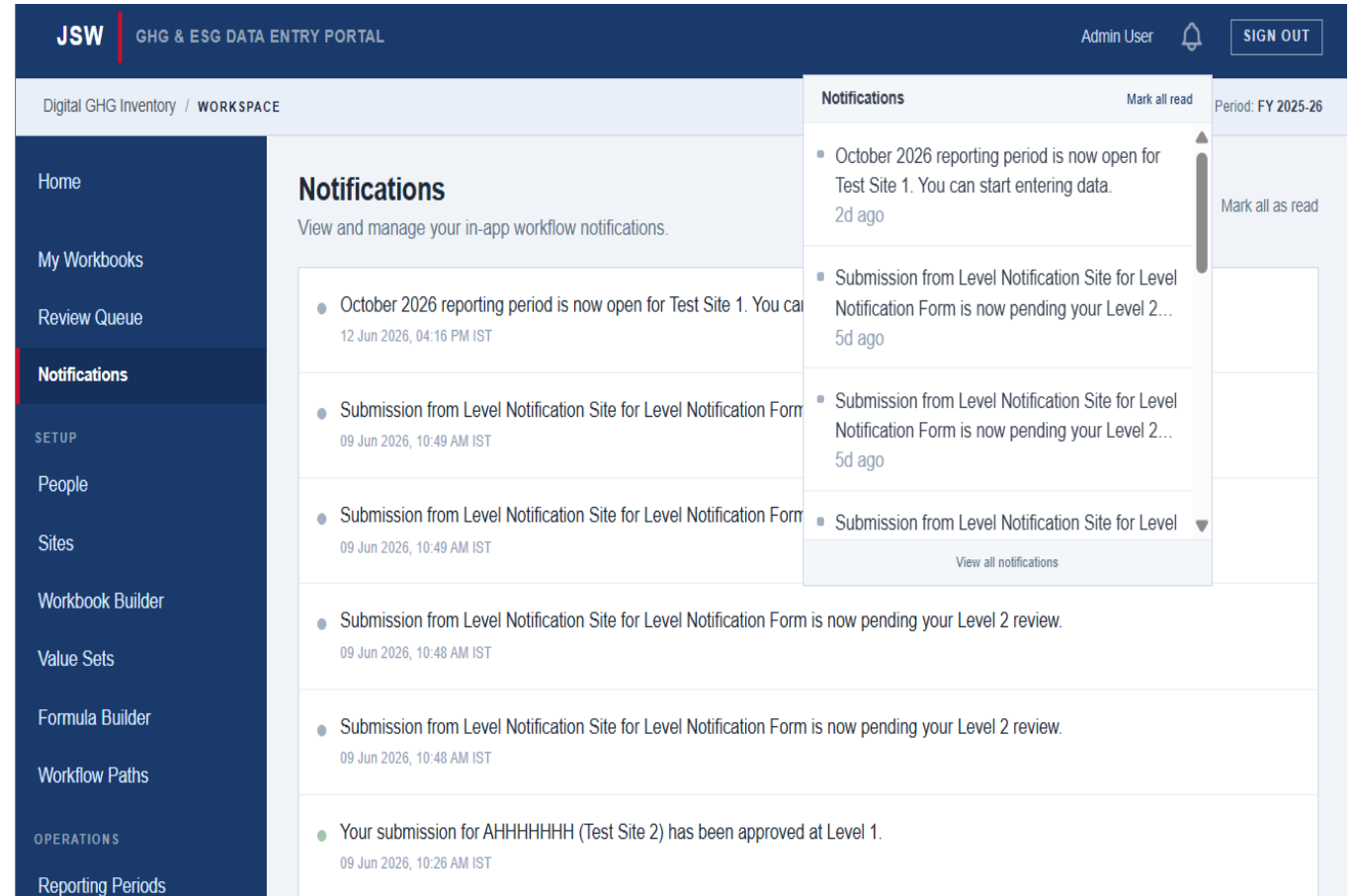
SHATAM RAI

Layer	Technologies Used	Purpose & Role in Project
Database	PostgreSQL, SQLAlchemy ORM, Alembic	Secure relational data storage for sheet values, sites, users, and audit logs. Schema versions are maintained dynamically via Alembic migrations.
Backend	Python 3.12, Flask, simpleeval	Core API and routing framework. The simpleeval package serves as the runtime mathematical evaluation engine for complex, user-defined GHG formulas.
Frontend	HTML5, Vanilla CSS, Tailwind CSS, Javascript (ES6+)	Interactive spreadsheet UI, auto-saving annual workbook, visual form builder, review route workflow canvas, and real-time dashboard feedback.
Utility & Security	openpyxl, reportlab, bcrypt	Parses/downloads Microsoft Excel templates, builds presentation reports in PDF format, and handles enterprise-grade password encryption.

MODULES IMPLEMENTED

Notification Engine

Dispatches event-driven in-app alerts to minimize reporting delays. It alerts SPOCs instantly when a sheet is sent back for corrections, feeds a navigation bell widget with direct links, and maintains a notification history dashboard for sustainability officers. Notifications are manually tested and work even on multi approval levels and change requests/rejections.

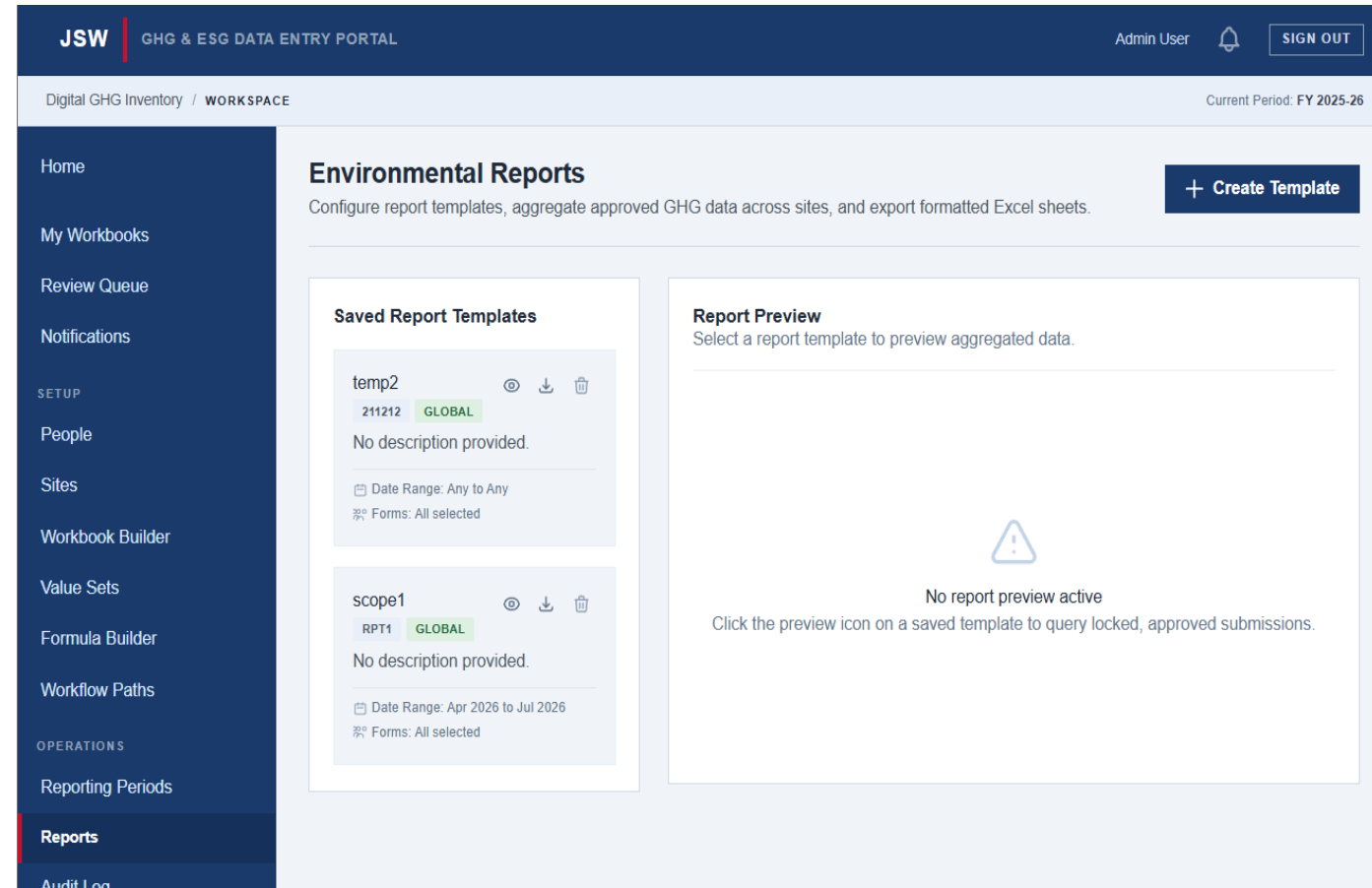


The screenshot displays the JSW GHG & ESG Data Entry Portal interface. The top navigation bar includes the JSW logo, the portal name, the user role (Admin User), a bell icon for notifications, and a SIGN OUT button. The main content area is titled "Digital GHG Inventory / WORKSPACE". On the left, a sidebar menu lists various modules: Home, My Workbooks, Review Queue, Notifications (highlighted), SETUP (People, Sites, Workbook Builder, Value Sets, Formula Builder, Workflow Paths), OPERATIONS (Reporting Periods), and a bell icon. The main content area shows a "Notifications" section with the heading "View and manage your in-app workflow notifications." Below this, a list of notifications is displayed, including: "October 2026 reporting period is now open for Test Site 1. You can start entering data." (12 Jun 2026, 04:16 PM IST), "Submission from Level Notification Site for Level Notification Form is now pending your Level 2 review." (09 Jun 2026, 10:49 AM IST), "Submission from Level Notification Site for Level Notification Form is now pending your Level 2 review." (09 Jun 2026, 10:48 AM IST), and "Your submission for AHHHHHHH (Test Site 2) has been approved at Level 1." (09 Jun 2026, 10:26 AM IST). A "Notifications" dropdown menu is open, showing a list of notifications and a "Mark all as read" button. The bottom right corner of the interface shows the period "FY 2025-26" and a "Mark all as read" button.

MODULES IMPLEMENTED

Report Builder

Aggregates activity data and resolves Scope 1, 2, and 3 emissions equations in a sandboxed calculation engine. It generates structured Excel sheets and landscape PDF reports for corporate disclosures, displaying real-time calculation previews alongside final approved reportable values.



MODULES IMPLEMENTED

Designing SPOC Sheets

The SPOC Sheet module is the main data entry interface where users enter monthly facility metrics. It presents an Excel-like grid styled with JSW Steel's color scheme, showing a white background and a blue left border on the month cell for the active reporting period. Month cells are color-coded to indicate their status, such as drafts or submitted sheets, using a dark font for high contrast. The tabs are arranged in a clean, scrollable single row with 8px status dots that show when a sheet is a draft or needs correction.

My Workbooks / Annual Workbook

Annual Workbook

Enter site data across the April-March financial year. Each month saves as a draft automatically.

Site

Test Site 1 (SITE1)

Financial year

FY 2026-27

Test Site 1

FY 2026-27

OPEN FOR ENTRY

Last saved 11 Jun 2026, 12:09 pm

MAKEININDIA

adsasdad

diesel

FORM3

NEWFORM

SECTIONTEST

FORMTEST2

Calculation Results

MONTH	NEW TEXT	NEW INTEGER	NEW NUMBER	REMARKS
April 2026	Navkar	24	12	—
May 2026	sadda	32	12	—
June 2026	443	12	3434	—
July 2026				
August 2026				
September 2026	—	—	—	—
October 2026				—
November 2026				
December 2026				
January 2027				
February 2027				
March 2027				

MODULES IMPLEMENTED

Designing Dashboard Widgets

As all archetypes (SPOC, Approver, Admin, superadmin etc.) in this application are treated as users, creating separate dashboards for different archetypes was nearly impossible. Hence dashboard was designed as an ensemble of different widgets which appear based on the permissions user has been given.

Example :

View Pending Approvals widget will only appear in the user who has approval permissions.

Setup and Monitoring

Readiness, open reporting months, and approved data.

SHEETS IN SCOPE

20

Sheets visible in your site scope

APPROVED AND LOCKED

4

Finalized reporting data

UNDER REVIEW

8

Monthly packages or sheets with reviewers

NEEDS ATTENTION

7

Sheets not yet approved for open months

Workspaces

People

Add and manage users who contribute, review, or manage reporting.

Sites

Manage ports and sites included in GHG reporting.

Workbook Builder

Build reusable workbook and sheet structures for site reporting.

Workflow Paths

Configure who reviews and approves each site monthly package.

Reports

View and export approved GHG data.

Recent Activity

Latest sheet activity relevant to your access.

SITE	SHEET NAME	LAST ACTION	STATUS
Level Notification Site	Level Notification Form	12 Jun 2026, 04:41 PM	DRAFT SAVED
Test Site 1	SECTIONTEST	12 Jun 2026, 04:18 PM	DRAFT SAVED
Test Site 1	MAKEININDIA	12 Jun 2026, 04:12 PM	DRAFT SAVED
Test Site 1	MAKEININDIA	12 Jun 2026, 01:02 PM	DRAFT SAVED
Test Site 1	MAKEININDIA	11 Jun 2026, 03:25 PM	SUBMITTED

Setup Progress

Core readiness for monthly GHG reporting.

People and access configured

Sites ready

Reporting months opened

Value Sets approved

Formula Builder rules published

Sheets configured

Workflow Paths assigned

Approved monthly package available